An electrochemical enzymatic biosensor based on *Xylaria* sp. laccase isolated from cassava waste is applied to quantify dopamine

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https://doi.org/10.1016/j.procbio.2024.12.002Get rights and content

Highlights

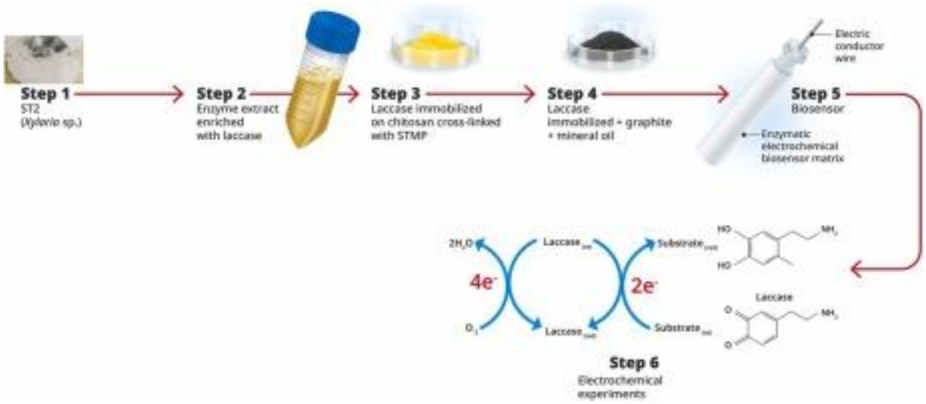
- A novel and sustainable *Xylaria* sp. laccase isolated from cassava waste.
- The laccase-based biosensor was successfully designed for dopamine quantification.
- Optimal parameters for dopamine quantification via cyclic voltammetry.
- Optimal laccase-based biosensor performance for dopamine quantification.

Abstract

Cassava waste (leaves and stems) was used to obtain endophytic fungi to assess laccase production. All the fungi (twenty-three) underwent an enzymatic extraction process to evaluate laccase production. The results revealed that six fungi from the stems and one from the leaves performed best. The fungus *Xylaria* sp. stood out for its greater productivity, and the growth parameters were optimized for this fungus (11 days, pH 6.70, and temperature 29°C). The enzyme extract enriched with laccase was then used to produce an electrochemical enzymatic biosensor immobilized in chitosan crosslinked with STMP. This biosensor was applied to quantify dopamine, with analysis conditions optimized for detecting low concentrations of the compound. The biosensor demonstrated its highest potential when operating at pH 5.60 and a temperature of 38.64°C, allowing the quantification of dopamine at concentrations from 0.17 to 492.83 μmol. L^{-1} with

a detection limit of 0.17 μmol. L^{-1} . The biosensor demonstrated high reproducibility (0.4 % error) and stability after seven cycles.

Graphical Abstract



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Introduction

Laccases, also known as polyphenol oxidases, belong to the class oxidoreductases. These enzymes are dimeric or tetrameric glycoproteins, typically containing four copper atoms per monomer or phenol oxidase [1], [2], [3], [4], [5], whose main source of obtaining is fungi, mainly white rot fungi. They can also be isolated from plants, insects, archaea, bacteria, and other fungi, mostly filamentous [6], [7], [8], [9], [10].

The laccases catalyze the oxidation of several compounds, including phenolic compounds, amines, polycyclic aromatic hydrocarbons, methoxy-substituted phenols, and certain inorganic compounds [11], [12]. The reaction occurs via phenoxyl radicals, reducing O_2 to H_2O through electron transfer [7], [13], hydrogen atom transfer, and ionic oxidation [14], [15]. Due to the diverse oxidation reactions catalyzed by laccases, they find applications in several industrial sectors, including cosmetic, textile, pharmaceutical, waste treatment, biofuel production, organic synthesis, pulp bleaching, paper production, and developing biosensors [4], [7], [16], [17], [18], [19], [20].

Among the various applications of laccases, studies involving the production of biosensors for quantification of phenolic compounds such as dopamine have been ongoing since the 1970s and have increasingly aroused the interest of researchers in recent years, especially concerning the production of enzymatic biosensors using oxidoreductases, including laccase. These studies are due to the need for advances in enzymatic biosensors since limitations in achieving high sensitivity at low concentrations of the analyzed compounds and using sustainable and economical materials for elaborating enzyme immobilization matrices to produce biosensors persist [3], [21], [22].

According to data extracted from Scopus databases using structured keywords, 104,791 studies were published on biosensors, including articles and reviews, with a significant increase in studies after 2010 (Fig. 1A). However, considering studies involving the application of biosensors using laccases, 698 studies were published (Fig. 1B), 104 of which involved laccases from *Trametes* species, six from *Agaricus*, eleven from *Rhus*, twenty-one from *Aspergillus*, nineteen from *Bacillus*, eleven from *Ganoderma*, four from *Pichia*, ten from *Cerrena*, twelve from *Pleurotus*, eight from *Pseudomonas*, and no studies involving the use of laccases from *Xylaria* (Fig. 1C). Few studies were also observed involving the production of laccase by *Xylaria* (only 26 studies), with most of the studies being applied in the remediation area.

When surveying the number of studies involving the production of specific enzymatic biosensors for dopamine quantification, 498 studies were highlighted, including articles and reviews, of which only 68 used laccases during their production, and none used the fungus *Xylaria* sp. as a source. Considering the production of laccases from fungi isolated from cassava residues, studies are even scarcer, with only one study being found. Therefore, the importance of applying agro-industrial residues as new sources of enzymes and the possibility of reusing these residues to obtain fungi, which are extremely promising for obtaining enzymes of technological interest, such as laccase, studied in the present work, is verified. It is also important to highlight that there is limited literature on the use of chitosan matrices, a very versatile, sustainable and economically viable biopolymer to produce biosensors, with only one article reporting amperometric biosensors to quantify dopamine and other phenolic compounds, such as pyrogallol, epicatechin, gallic acid, 1,2-dihydroxybenzoic acid, caffeic acid, chlorogenic acid, rutin and catechin [23].

The limited number of published studies highlights the need to advance further in developing additional biocomponents using this biopolymer, which will enable the reduction of device costs and greater applicability on a large scale. Therefore, there is a need for advances in studies involving new sources of enzymes such as laccases. It should also be taken into account that dopamine, a catecholamine with essential biological functions as a hormone and neurotransmitter linked to the central nervous, endocrine, and cardiovascular systems, serves as a marker for several diseases, such as schizophrenia, dementia, Alzheimer's, Parkinson's and HIV infection, which shows the importance of performing rapid and efficient quantifications of it in body fluids as proposed in this research since accurate quantification of dopamine plays a fundamental role in these disorders and early and fast detection can significantly impact patient outcomes [24], [25],

[26]. Therefore, considering the importance of exploring new sources of laccases and developing new devices capable of quantifying neurotransmitters such as dopamine at low concentrations, this study presents a novel approach that combines sustainability with biotechnological innovation since cassava residues were used as a substrate to cultivate endophytic fungi capable of producing the laccase enzyme.

Unlike previous works focused on traditional fungal species or commercial sources of enzymes, this research explores the potential of *Xylaria* sp., isolated from cassava residues, to produce enzymes to develop biosensors. Furthermore, the production of biosensors exclusively employs a matrix based on chitosan crosslinked with STMP, which has demonstrated high enzyme immobilization efficiency and stability in detecting dopamine at low concentrations. Therefore, this new methodology establishes a basis for developing cost-effective and scalable biosensors applicable in medical diagnostics and environmental monitoring, marking a significant contribution to biosensor technology and waste valorization.

Materials

Endophytic fungi were isolated from stems and leaves of cassava waste and maintained at FEI. Potato dextrose agar (PDA), chitosan (low molecular weight), syringaldazine, malt extract, glucose, peptone, serum ovalbumin, ammonium sulfate, graphite (100 mesh), Tween 80, trisodium trimetaphosphate (STMP), polyvinylpolypyrrolidone (PVPP), coomassie blue, *L*-dopamine, gentamicin sulfate, sodium hydroxide pellets (≥ 98 %), acetic acid, sodium acetate, copper sulfate, potassium phosphate monobasic,

Isolation of endophytic fungi and obtained of enzymatic extracts

Pure fungal lineages associated with cassava leaves (LV) and stems (ST) were successfully isolated. Table 1 identifies eighteen fungi from the stems and five from the leaves. Six fungi derived from the stems (ST1 – *Microdochium lycopodinum*, ST6 - *Phyllosticta elongata*, ST9 - *Diaporthe endophytica*, ST12 - *Diaporthe caatingaensis*, ST13 - *Stenocarpella maydis*, and ST15 - *Sordariomycetes* sp) and one fungus from the leaves (LV2 - *Xylaria* sp.) displayed significant laccase activity.

Fungi ST1, ST15,

Conclusion

It was possible to isolate and identify endophytic fungi associated with cassava waste, demonstrating their potential for producing enzymes for biotechnological applications, mainly LV2 (*Xylaria* sp.), which showed the highest laccase production, which was immobilized in chitosan and applied in the production of electrochemical enzymatic biosensor. The resulting biosensor exhibited excellent performance in dopamine quantification with sensitivity up to 0.30 μmol.L^{-1}.

These results highlight the

CRediT authorship contribution statement

Andreia de Araújo Morandim-Giannetti: Writing – original draft, Validation, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Vitor Stabile Garcia:** Methodology, Investigation, Formal analysis. **Gerhard Ett:** Writing – review & editing, Writing – original draft, Validation, Data curation, Conceptualization. **Patrícia Alessandra Bersanetti:** Methodology, Investigation, Data curation. **Mariana Pontes Vieira:** Methodology, Investigation,

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Andreia de Araujo Morandim-Giannetti reports financial support was provided by State of Sao Paulo Research Foundation. Mariana Pontes Vieira reports financial support was provided by State of Sao Paulo Research Foundation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have

Acknowledgments

The authors acknowledge the São Paulo Research Foundation (FAPESP) (Grant Numbers #2015/19273–2 and #2022/09183–0) for providing financial support to carry out the research work and Alex Argozino for elaborating Fig. 5 and the graphical abstract.